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# Analyzing Loan Approvals through Supervised Machine Learning Techniques

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**Abstract— The main goal of this research work is to apply supervised machine learning algorithms to forecast loan acceptance. The data-set comprises 500 instances with 13 features including socioeconomic attributes and credit history of loan applicants, with the target variable indicating loan approval status. Twelve distinct supervised machine learning algorithms are the subjects of performance comparison. The models are evaluated based on test accuracy scores, precision and F1 Score. Furthermore, utilizing visualization methods like confusion matrices and heat maps to acquire adequate understanding of the model's performance and pinpoint possible areas for enhancement. The test results offer insightful information on how well different machine learning algorithms predict loan acceptance, thereby aiding financial institutions in making informed decisions and mitigating risks associated with lending practices.**

***Keywords— Classification Models, Credit Risk Assessment, Loan Approval, Predictive Analysis, Supervised Machine Learning***

## I. INTRODUCTION

The banking sector faces persistent challenges, especially concerning loan approval mechanisms susceptible to fraudulent activities. Instances of fraud documentation, signature forgery, and numerical alterations pose significant financial risks, leading to substantial losses for financial institutions. These illicit practices underscore the urgent need for robust predictive models to enhance fraud detection and effectively mitigate the associated risks [1]. The introduction of data analytics and machine learning has revolutionized the finance industry, particularly in the domain of banking and loans. With the proliferation of digital data and technological advancements, financial institutions now leverage sophisticated algorithms to analyze vast datasets, enabling accurate risk assessment and decision-making processes.

In this research, the difficulties in loan permitting processes are analyzed and are addressed through the application of supervised machine learning algorithms. Using Python programming language and its comprehensive libraries for machine learning, including ski-kit-learn, the test data can explore a diverse range of supervised classification models. These models encompass several different algorithms, including Support Vector Machine (SVM), Random Forest, Logistic Regression, and Decision Trees, each offering unique capabilities in predicting loan approval outcomes [2]. In order to assess and display the outcome of the machine learning models that are included in the use of advanced techniques such as seaborn for creating heat maps and matplotlib for generating confusion matrices are displayed. The use of visualization tools gains deeper insights into the predictive accuracy and efficacy of the models, facilitating informed decision-making for financial institutions. Through this project, the aim is to demonstrate the practical application of machine learning in strengthening the effectiveness and dependability of loan approval procedures, thus helping the general integrity and stability of the banking industry.

## II. LITERATURE REVIEW

The literature review on loan acceptance with the use of machine learning algorithms to forecast outcomes exposes a growing body of research exploring various methodologies and approaches to improve the accuracy as well as effectiveness of lending decisions in the banking sector [3]. Several studies have investigated the application of supervised machine learning techniques such as Random Forest, Support Vector Machine (SVM), and Logistic Regression in predicting loan approval outcomes based on borrower characteristics, credit history, and financial indicators [4].

Researchers have highlighted the significance of feature selection and preprocessing techniques to improve model performance and reduce the dimensionality of the dataset. Additionally, ensemble methods such as Gradient Boosting and AdaBoost have been explored for their ability to combine multiple weak learners to produce stronger predictive models [5].
Moreover, studies have emphasized the importance of evaluating the performance of models using suitable metrics, such as recall, efficiency, precision and F1 score [6], while also considering potential trade-offs between false positives and true positives rates in loan approval decisions.

Overall, the literature underscores the potential algorithms of machine learning to optimize the loan approval method, enabling financial institutions to decide on loans more sensibly and consistently while minimizing risks associated with default and fraud [7].

## III. METHODOLOGY

This phase describes the Dataset used, Preprocessing, experimental settings, model evaluation techniques, Machine Learning models, and statistical techniques carried out for the research.

### *A. Dataset*

The collected data consists of twelve independent variables and one dependent variable. The independent variable contains the various data associated with different loan applicants. The dependent variable, labeled "Loan_Status", depicts the approval of loans. This lists the various criteria for the approval of loans.

The dataset contains 12 different categories. The top categories in the list of attributes are data for Gender, Marital Status, Educational Qualification, Number of Dependents, Self-employment Status, Applicant Income, Co-applicant Income, Loan Amount, Loan period, Credit History, Property Area, Loan Status.

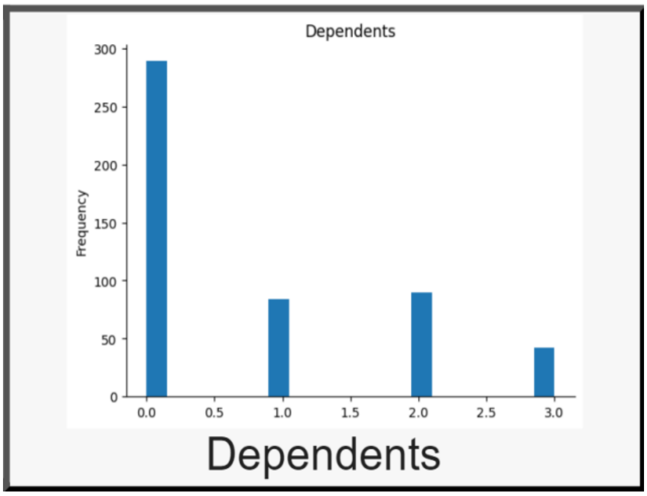


Fig. 1. Bar-chart representation of Number of Dependents

Three bar charts are plotted to show the frequency and count of applicants on the basis of approval status for Number of Dependency, Basis of maternal Status and Educational Qualification. In the used dataset 280 applicants have no Dependents, allowing their chances of approval is higher. Whereas it is observed the number of dependents as 1, 2 and 3+ gradually decreased chances of approval status.

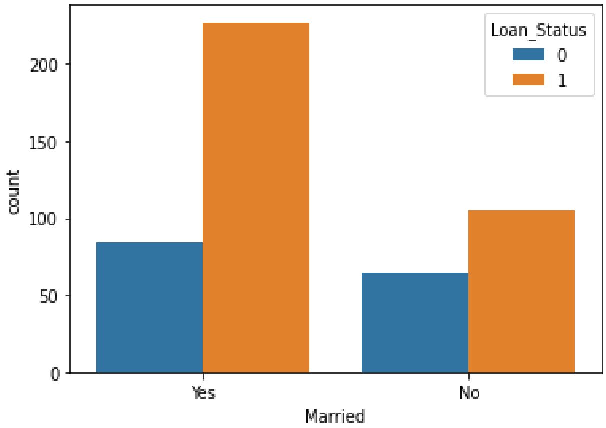


Fig. 2. Bar-chart representation of Marital Status

Marital status “fig. 2” plays a huge role in the loan approval process. Married applicants who have some amount of co-applicant income have a higher chance of getting approved by the bank. Whereas a complete dependent situation may lead to the rejection of the credit.

In case of unmarried applicants, other instances are taken into consideration like their educational qualification, annual income and credit history.

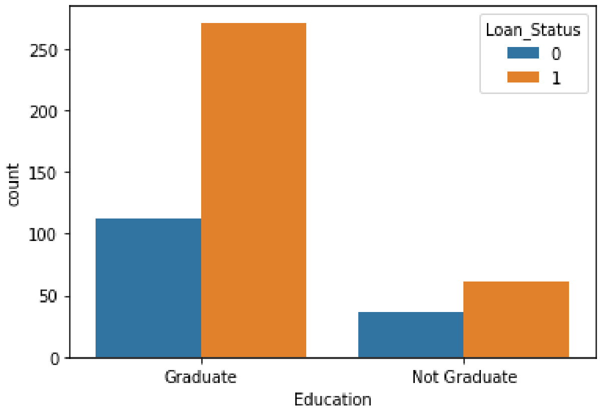


Fig. 3. Bar-Chart representation of Education Qualification

Bar-graphs in “fig. 3” show the impact of educational qualification on the approval status of the candidate’s loan. An applicant with graduation has a higher chance of success to get the credit amount compared to a non-graduate candidate.

### *B. Data Preprocessing*

The dataset contains CSV files of multiple instances. The Loan ID contains unnecessary and unprocessed information; this needs to be dropped before feeding into the ML model. Preprocessing and preparing data for predicting outcomes requires a significant amount of effort. During data preprocessing, less informative columns were removed, NULL values were removed and data is standardized [8]. The data in attributes like Education, Married, Loan Status and property area were encoded. ‘Yes’ equals ‘1’ and ‘No’ equals ‘0’.

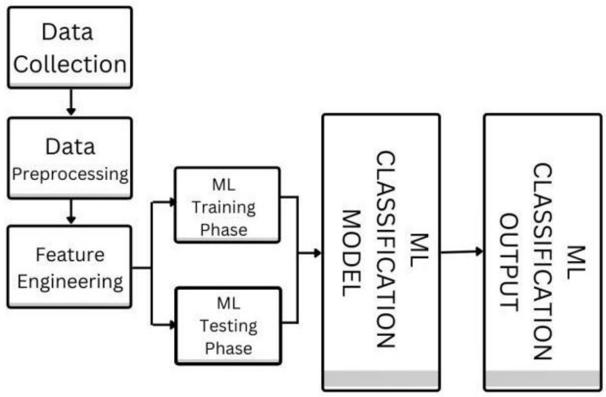


Fig. 4. Block diagram showing all stages of the study

The data pre-processing component cleans and prepares the resume data for machine learning [7].
This includes the following steps:

1. *Removing the NULL Values:* NULL Values Such as “NaN”, blank spaces need to be removed. The machine learning model's accuracy is improved by removing these NULL values from the dataset.

2. *Using the mean values:* The other way to handle NULL values is to fill up the empty data cells with the average of all the values present for the attribute.

3. *Encoding:* Process of encoding a data is to give “1” or “0” for getting better results from the predictive machine. For example, “Yes” can be encoded as “1” and “No can be encoded as “0”.

4. *Standardization:* Standardization before fitting data in a model is crucial as it ensures all features are on a similar scale, preventing bias towards variables with larger magnitudes. It improves convergence speed in optimization algorithms like Gradient Descent, enhances model performance by reducing the impact of outlier, and facilitates interpretability and comparability across different models [9].

### *C. Correlation*

Identifying correlation between the attributes present in the dataset is a must-do practice. The correlations define the strong or weak relation between the attributes. Seaborn’s heatmap is used to generate the correlation matrix.

### *D. Machine Learning Models*

The original dataset is split into 80% for the training of models and the rest 20% for the testing sets, respectively. The input features from train data were used for model training on several classifiers after train/test splits on the original data set [9]. Using the test data, the forecast was created, and the accuracy metrics were recorded.

*Logistic Regression:* Using a threshold value, the logistic function is applied to the classification job in logistic regression. It is regarded as one of the simplest classification problem implementations.

*Decision Tree Classifier:* Any internal node in a decision tree classifier represents an attribute test, any branch indicates the test's result, and any leaf node offers the final prediction. Once built, the decision tree classifier may predict new data by iteratively navigating the branches of the tree that match to the characteristic values in the data [10].

*Random Forest Classifier:* The Random Forests Classifier picks multiple samples from the training dataset (with replacement) and independently trains the model for each sampled dataset. The average of all forecasts from all sub models is used to predict the final result. Although it should be emphasized that the trees are formed in a way that lowers the relationship between individual classifiers, the samples from the training dataset are gathered without replacement. For every split, an arbitrary subset of features is taken into account, rather than depending on the greedy selection of the ideal split point during the construction of each tree.

*KNN Classification:* KNN (K-Nearest Neighbors) Classification is an algorithm of machine learning from the instance-based learning family. A new data point is classified using this non-parametric method by comparing it to its nearest neighbor in the training dataset [11].

*Support Vector Machine (SVM):* A supervised learning model that can be implemented on tasks involving regression and classification. The support vectors—the closest data points—are utilized to depict the different categories; they function as the boundary that optimizes the space between them [12].

*Naive Bayes:* An algorithm that uses probability to produce forecasts about the chances that particular events will take place. It learns the likelihood of different variables first, considering the different job categories. The Bayes theorem is then applied to ascertain the job category to which a cleaned resume belongs [13].
It has two main advantages: it is simple and efficient because it requires relatively little training data and can be taught quickly.

This makes it suitable for data with a variety of features and speeds up the process of computing probability.
Naive Bayes Classifiers are of 3 types:
1. *Multinomial Naive Bayes (MNB)*- Assumes that features are generated from a multinomial distribution.
2. *Bernoulli Naive Bayes (BNB)*- Assumes that features are binary variables.
3. *Gaussian Naive Bayes (GNB)*- It forecasts a Gaussian distribution for continuous features.

In this study, The Gaussian Naive Bayes Classifier for classification task is used, due to better accuracy in such case

### *E. Plotting the Result*

Confusion matrix is the best choice to display the results of a machine learning algorithm in case of a classification type problem.

## IV. IMPLEMENTATION AND RESULTS

The outcomes of the prediction models are discussed in this section.

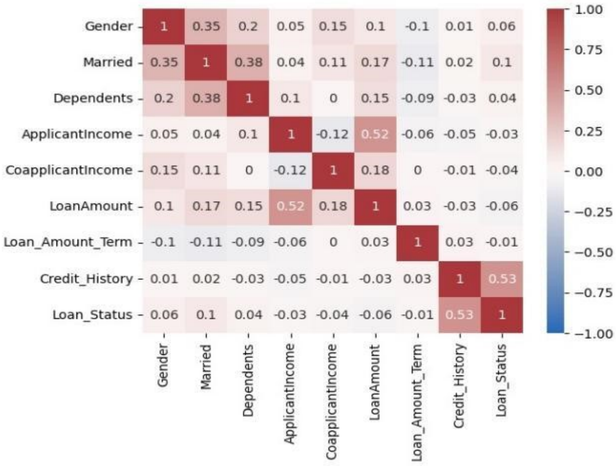


Fig. 5. Correlation matrix showing relation among all attributes

Correlation matrix, visualizes the different strengths and weaknesses of relations between two different features in a dataset. Where, values close to 1 and -1 represent the stronger correlation between the two features, whereas values close to zero define a weaker correlation. In this correlation matrix "Fig.5", shows the relation between 9 different features of the dataset to deduce the approval status of the applicant's loan.

*Accuracy:* Facilitates understanding of the model's capacity to forecast true positive and true negative classifications correctly.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

*Precision:* This statistic helps evaluate the reliability of the model.

$$Precision = \frac{TP}{TP + FP}$$

*Recall:* A performance statistic that assesses how well a classifier can identify every positive occurrence in the dataset among all of the real positive examples.

$$Recall = \frac{TP}{TP + FN}$$

*F1-Score:* The F1-score gives a fair evaluation of a classifier's performance. This combines precision and recall into a single number. It is the harmonic mean of precision and recall, giving each metric the same amount of weight. [14]

$$\text{F1-Score} = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

These metrics are all summarized in Tables II and III for each of our classification models, applied to both natural language processing techniques.

TABLE I. Metrics for Negative Class

| **Performance Metric for Negative Class** | | | | |
|---|---|---|---|---|
| ***Model*** | ***Precision*** | ***Recall*** | ***F1-Score*** | ***Accuracy*** |
| Random Forest Classifier | 0.80 | 0.53 | 0.64 | 82.17 |
| KNeighbors Classifier | 0.84 | 0.53 | 0.65 | 83.16 |
| SVM(Support-vector Machine) | 1.00 | 0.47 | 0.64 | 84.15 |
| Logistic Regression | 1.00 | 0.47 | 0.64 | 84.15 |
| Decision Tree Classifier | 0.50 | 0.50 | 0.50 | 70.83 |
| GaussianNB | 0.82 | 0.47 | 0.60 | 81.18 |
| XGB Classifier | 0.69 | 0.67 | 0.68 | 81.18 |
| Gradient-Boosting Classifier | 0.78 | 0.47 | 0.58 | 80.19 |
| AdaBoost Classifier | 0.83 | 0.50 | 0.62 | 82.17 |
| Multi-linear Perceptron (Neural-network) | 0.82 | 0.47 | 0.60 | 81.18 |
| Bagging Classifier | 1.00 | 0.47 | 0.64 | 84.15 |
| Stacking Classifier | 0.92 | 0.40 | 0.56 | 81.18 |

TABLE II. Metrics for Positive Class

| **Performance Metric for Positive Class** | | | | |
|---|---|---|---|---|
| ***Model*** | ***Precision*** | ***Recall*** | ***F1-Score*** | ***Accuracy*** |
| Random Forest Classifier | 0.83 | 0.94 | 0.88 | 82.17 |
| KNeighbors Classifier | 0.83 | 0.96 | 0.89 | 83.16 |
| SVM(Support-vector Machine) | 0.82 | 1.00 | 0.90 | 84.15 |
| Logistic Regression | 0.82 | 1.00 | 0.90 | 84.15 |
| Decision Tree Classifier | 0.79 | 0.79 | 0.79 | 70.83 |
| GaussianNB | 0.81 | 0.96 | 0.88 | 81.18 |
| XGB Classifier | 0.86 | 0.87 | 0.87 | 81.18 |
| Gradient-Boosting Classifier | 0.81 | 0.94 | 0.87 | 80.19 |
| AdaBoost Classifier | 0.82 | 0.96 | 0.88 | 82.17 |
| Multi-linear Perceptron (Neural-network) | 0.81 | 0.96 | 0.88 | 81.18 |
| Bagging Classifier | 0.82 | 1.00 | 0.90 | 84.15 |
| Stacking Classifier | 0.80 | 0.99 | 0.88 | 81.18 |

The Best accuracy is by the SVC and Logistic Regression. Accuracy score is 84.15%. Listed below are the results of the best classification technique, in the form of confusion matrix [15].

Figure 8 and Figure 9 show the confusion matrix for the SVC and Logistic Regression. These confusion matrices are produced by using the Gaussian Naive Bayes model on the dataset for prediction.

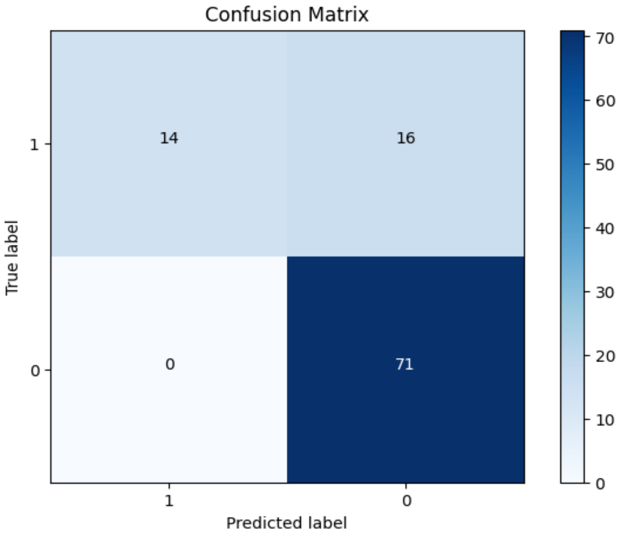


Fig. 8. Confusion matrix of Support-Vector Classifier (SVC)

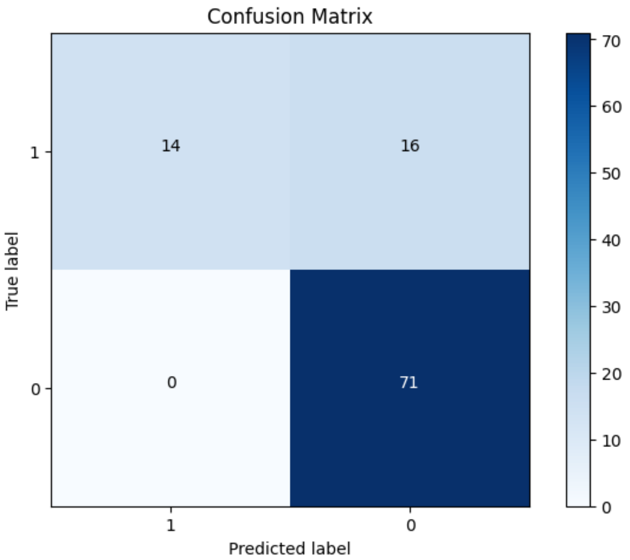


Fig. 9. Confusion matrix for Logistic Regression

This is the reason that recall values for the positive class are really close to 1. Hence, these two models can be used for ML engineers to develop any app for approval of loan status as this is more reliable.

Since most of the values in the Loan Status column were ‘0’, that is Not approved by the bank. The Negative numbers are higher. The best fit models show, True positive 14, True negative 71 and False positive 16, False negative 0.

TABLE III. Loss History of the Keras Model

| **Training Loss Value** | **Testing Loss Value** |
|---|---|
| [1.2470223903656006,<br>1.0752180814743042,<br>0.952262818813324,<br>0.858955442905426,<br>0.7877222299575806,<br>0.733892142772674 6,<br>0.6911346912384033,<br>0.6575238108634949,<br>0.6309635043144226,<br>0.6085962653160095] | [0.5828315019607544,<br>0.574843168258667,<br>0.566132664680481,<br>0.557370126247406,<br>0.548673450946807 9,<br>0.5404345989227295,<br>0.5318412184715271,<br>0.5246325135231018,<br>0.5174928903579712,<br>0.5087908506393433] |

The performance of keras model on the basis of Loss Function and Accuracy is recorded in the “Table III” and “Table IV” respectively. The loss value greatly reduces from 1.247 to 0.608 and furthermore it reduces from 0.582 to 0.508 for the test cases.

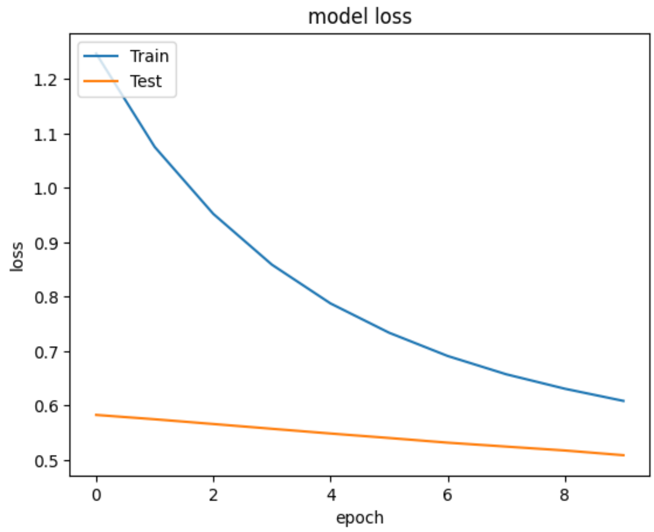


Fig. 10. Loss curve of the Keras Model

The loss curve “Fig 10” and the accuracy curve “Fig 11” shows the model’s performance in the graphical manner.

The accuracy of the model also increased gradually from 0.418 to finally 0.831 throughout 10 epochs and a batch size of 10. This successfully assures the predictive neural network is neither underfitting nor overfitting for this data set.

TABLE IV. Accuracy History of the Keras Model

| **Training accuracy value** | **Testing accuracy value** |
|---|---|
| [0.4183168411254883,<br>0.5297029614448547,<br>0.5965346693992615,<br>0.6435643434524536,<br>0.6831682920455933,<br>0.7103960514068604,<br>0.7425742745399475,<br>0.7722772359848022,<br>0.7846534848213196,<br>0.7945544719696045] | [0.8217821717262268,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117,<br>0.8316831588745117] |

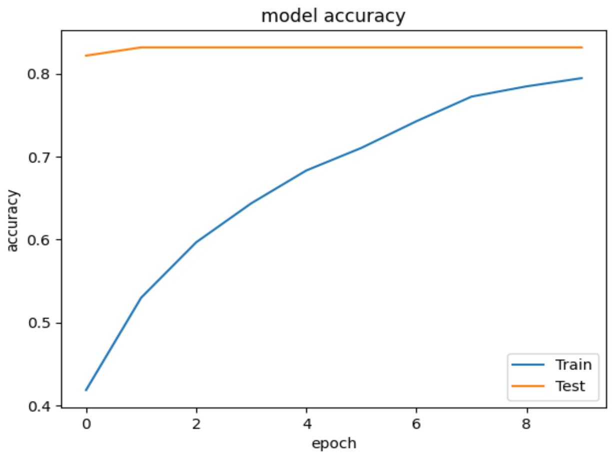


Fig. 11. Accuracy curve of Keras Model

## V. CONCLUSION

The results indicate that while the Support Vector Machine (SVM) outperforms other models in overall accuracy and recall for loan approval prediction, different models exhibit varying strengths across different evaluation metrics. Logistic Regression, Gradient Boosting, and Stacking Classifier also demonstrate competitive performance, suggesting their potential utility in real-world applications. The nuanced differences in model performance underscore the importance of comprehensive evaluation and the need to consider multiple metrics when selecting the most suitable approach for loan approval prediction. All things considered, this study demonstrates how effective supervised machine learning algorithms can be in raising the effectiveness and dependability of loan approval procedures in the banking industry. This research study lays the foundation for further advancements in the field of loan approval prediction using machine learning. Future endeavors could explore ensemble techniques to combine the strengths of multiple models and improve predictive accuracy.

## REFERENCES

[1] Sheikh, Mohammad Ahmad, et al. "An Approach for Prediction of Loan Approval Using Machine Learning Algorithm." 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC), 2020, pp. 490–94. IEEE Xplore, https://doi.org/10.1109/ICESC48915.2020.9155614.

[2] Saini, Prabaljeet Singh, et al. "Loan Approval Prediction Using Machine Learning: A Comparative Analysis of Classification Algorithms." 2023 3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), 2023, pp. 1821–26. IEEE Xplore, https://doi.org/10.1109/ICACITE57410.2023.10182799.

[3] Lik Pao, Paul Law, and Mohd Arfian Ismail. "Loan Eligibility Classification Using Logistic Regression." 2023 IEEE 8th International Conference On Software Engineering and Computer Systems (ICSECS), 2023, pp. 1–4. IEEE Xplore, https://doi.org/10.1109/ICSECS58457.2023.10256402.

[4] Rahman, Arifa Tur, et al. "Prediction of the Approval of Bank Loans Using Various Machine Learning Algorithms." 2023 IEEE World Conference on Applied Intelligence and Computing (AIC), 2023, pp. 272–77. IEEE Xplore, https://doi.org/10.1109/AIC57670.2023.10263880.

[5] Shaheen, Salma Khaled, and Essam ElFakharany. "Predictive Analytics for Loan Default in Banking Sector Using Machine Learning Techniques." 2018 28th International Conference on Computer Theory and Applications (ICCTA), 2018, pp. 66–71. IEEE Xplore, https://doi.org/10.1109/ICCTA45985.2018.9499147.

[6] Sun, Hui, and Mingyuan Guo. "Credit Risk Assessment Model of Small and Medium-Sized Enterprise Based on Logistic Regression." 2015 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), 2015, pp. 1714–17. IEEE Xplore, https://doi.org/10.1109/IEEM.2015.7385940.

[7] Wang, Qiang, et al. "Green Credit Scoring System and Its Risk Assessemt Model with Support Vector Machine." 2011 Fourth International Joint Conference on Computational Sciences and Optimization, 2011, pp. 284–87. IEEE Xplore, https://doi.org/10.1109/CSO.2011.143.

[8] Changjian, Li, and Hu Peng. "Credit Risk Assessment for Rural Credit Cooperatives Based on Improved Neural Network." 2017 International Conference on Smart Grid and Electrical Automation (ICSGEA), 2017, pp. 227–30. IEEE Xplore, https://doi.org/10.1109/ICSGEA.2017.161.

[9] Prasanth, C., et al. "Intelligent Loan Eligibility and Approval System Based on Random Forest Algorithm Using Machine Learning." 2023 International Conference on Innovative Data Communication Technologies and Application (ICIDCA), 2023, pp. 84–88. IEEE Xplore, https://doi.org/10.1109/ICIDCA56705.2023.10100225.

[10] *Singh, Vishal, et al. "Prediction of Modernized Loan Approval System Based on Machine Learning Approach." 2021 International Conference on Intelligent Technologies (CONIT), 2021, pp. 1–4. IEEE Xplore,*

[11] Shrivastava, Abhishek, et al. "A Novel Hybrid Model for Predictive Analysis of Myocardial Infarction Using Advanced Machine Learning Techniques." 2023 OITS International Conference on Information Technology (OCIT), 2023, pp. 381–86. IEEE Xplore, https://doi.org/10.1109/OCIT59427.2023.10430780.

[12] Loveless, T. D., et al. "Analysis of Single-Event Transients (SETs) Using Machine Learning (ML) and Ionizing Radiation Effects Spectroscopy (IRES)." IEEE Transactions on Nuclear Science, vol. 68, no. 8, Aug. 2021, pp. 1600–06. IEEE Xplore, https://doi.org/10.1109/TNS.2021.3050879.

[13] Vijayalakshmi, M., et al. "Predictive Modeling of Cardiovascular Disease Using ML Algorithms." 2023 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI), 2023, pp. 1–7. IEEE Xplore, https://doi.org/10.1109/ICDSAAI59313.2023.10452498.

[14] Panda, Amiya Ranjan, et al. "Ad Click-Through Rate Prediction: A Comparative Study of Machine Learning Models." 2024 International Conference on Emerging Systems and Intelligent Computing (ESIC), 2024, pp. 679–84. IEEE Xplore, https://doi.org/10.1109/ESIC60604.2024.10481562.

[15] Panda, Amiya Ranjan, et al. "Classification of Customer Churning Based on OTT Platform Data." 2024 International Conference on Emerging Systems and Intelligent Computing (ESIC), 2024, pp. 673–78. IEEE Xplore, https://doi.org/10.1109/ESIC60604.2024.10481640.